ClaimFast: Motor Insurance Claim Triage & Entity Extraction

Rigorous Benchmark Evaluation of Prompt Engineering Strategies on High-Noise Indian Claims

Model: qwen/qwen3.8-27b (Groq Cloud) | Temperature: 0.0 | Output: Strict JSON Object Mode | 900 Jobs

1. Executive Summary & Experimental Mandate

Motor insurance First Notice of Loss (FNOL) customer communications in India are characterized by high linguistic noise, code-mixed Hinglish phrases, ambiguous damage descriptions, and missing identifiers. The **ClaimFast** benchmark evaluates the capability of open-weights LLMs to reliably parse unstructured customer messages into a structured 9-field schema, disambiguate complex insurance legal boundaries, and extract actionable triage metadata without external tool intervention.

The evaluation was executed across **900 planned jobs** representing 60 held-out test messages evaluated across 5 prompt strategies over 3 repeated stochastic runs at temperature 0.0. Exactly 900 unique job records were completed with **0 missing and 0 duplicate jobs**, achieving a **99.44% execution success rate** (895 / 900).

Strategic Pillar	Key Metric	Empirical Finding & Significance
Safety Triage	High-Urgency F1: 1.0000	**100.0% Precision & Recall** across all 240 high-urgency instances. Zero safety escapes.
Category Triage	Macro-F1: 0.9844	100% precision on theft & total loss; 98.1% overall accuracy across all 6 claim taxonomy classes.
Extraction Fidelity	Policy/Date Acc: 100%	Zero hallucinations on policy numbers, ISO dates, vehicle registrations, and claim amounts.
Optimal Strategy	Structured Reasoning	Highest classification Macro-F1 (0.9870), lowest latency (510ms), 24% faster than few-shot k=8.
Full Record Match	Few-Shot k=8: 86.1%	In-context span demonstration elevated strict 9-field exact match from 61.7% to 86.1%.

2. Dataset Architecture & Evaluation Matrix

The benchmark dataset was partitioned into 20 development messages (data/dev.jsonl) used exclusively for few-shot exemplar selection, and 60 strictly held-out test messages (data/test.jsonl) used for final scoring. All messages reflect authentic Indian driving conditions, including municipal towing non-claims, hit-and-run incidents, constructive total loss assessments, component theft, and Hinglish transliterations.

Dataset Split	Message Count	Usage in Experiment	Integrity & Separation Control
Development (dev.jsonl)	20 records	Exemplar source for k=3, k=5, k=8	Strictly excluded from evaluation runner scoring.
Held-Out Test (test.jsonl)	60 records	Final evaluation scoring benchmark	Zero exemplar contamination in any prompt template.
Total Planned Matrix	900 jobs	60 records × 5 strategies × 3 runs	100% deterministic tracking with unique composite job IDs.

3. Controlled Prompt Engineering Strategies

To isolate the impact of in-context demonstration scale versus explicit cognitive deliberative checkpoints, five controlled prompt architectures were evaluated against identical messages:

Strategy	k	Tokens	Architectural Hypothesis & Design Specification
zero_shot	0	1,247	Baseline extraction relying purely on detailed system definitions and schema specifications.
few_shot_k3	3	1,748	Minimal demonstrative grounding for third-party liability, own-damage, and glass damage.
few_shot_k5	5	2,112	Adds component theft (Rule A) and administrative towing non-claims.
few_shot_k8	8	2,712	Full coverage: constructive total loss, untraced hit-and-run, and Hinglish injury.
structured_reasoning	0	1,817	Internal verification checklist (Rule A/B/C deliberation) emitting only raw JSON without chain-of-thought.

4. Benchmark Performance & Evaluation Matrix

All metrics below reflect the complete 900-job execution matrix (180 runs per strategy). Non-success jobs are penalized as zero-score mismatches across all fields, ensuring absolute statistical rigor without cherry-picking.

Metric	Zero-Shot	Few-Shot k=3	Few-Shot k=5	Few-Shot k=8	Structured Reas.	Benchmark
Sample Count (N)	180	180	180	180	180	**900**
Valid Schema Rate	98.3%	99.4%	**100.0%**	99.4%*	**100.0%**	**99.4%**
Strict Match (9 Fields)	61.7%	64.4%	71.7%	**86.1%**	65.0%	**69.8%**
Normalized Match (Loc Norm)	78.3%	81.1%	80.0%	**86.1%**	81.7%	**81.4%**
Claim Type Accuracy	98.3%	97.8%	98.3%	97.8%	**98.3%**	**98.1%**
Claim Type Macro-F1	0.9851	0.9822	0.9851	0.9827	**0.9870**	**0.9844**
Urgency Accuracy	91.1%	**92.8%**	91.7%	**92.8%**	91.7%	**92.0%**
Urgency Macro-F1	0.9057	**0.9305**	0.9143	0.9283	0.9143	**0.9188**
Policy Number Acc	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
Incident Date Acc	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
Location Acc (Strict)	73.3%	73.3%	81.7%	**92.8%**	73.3%	**78.9%**
Location Acc (Normalized)	90.0%	90.0%	90.0%	**92.8%**	90.0%	**90.6%**
Vehicle Reg No Acc	100.0%	100.0%	100.0%	99.4%*	100.0%	**99.9%**
Estimated Amount Acc	99.4%	100.0%	100.0%	99.4%*	100.0%	**99.8%**
Injury Reported Acc	99.4%	100.0%	100.0%	99.4%*	100.0%	**99.8%**
Police Report Filed Acc	100.0%	100.0%	100.0%	99.4%*	100.0%	**99.9%**
Mean Latency (ms)	549.0	545.9	563.7	668.9	**510.0**	**567.4**
Median Latency (ms)	502.0	516.0	549.3	585.4	**483.0**	**528.5**
P95 Latency (ms)	863.6	774.4	790.9	1314.6	**662.8**	**811.4**
Mean Input Tokens	1,247	1,748	2,112	2,712	1,817	**1,926**
Cost / 1k Claims ($)	$0.2709	$0.3711	$0.4439	$0.5641	$0.3849	**$0.4070**

**Few-Shot k=8 valid schema rate and field accuracies reflect 179 successfully executed jobs and 1 API token rate limit error scored as 0.*

5. Safety-Critical Triage: Injury & High-Urgency Evaluation

In insurance operational automation, misclassifying bodily injury or undrivable vehicles as low urgency creates critical legal and human safety liabilities. The evaluation audited every prompt strategy separately for both **High Urgency** (16 gold records × 3 runs = 48 target instances) and **Injury Reported** (8 gold records × 3 runs = 24 target instances):

Strategy	Target Field	TP	FP	FN	TN	Precision	Recall	F1-Score
zero_shot	High Urgency	48	0	0	132	100.0%	100.0%	**1.0000**
few_shot_k3	High Urgency	48	0	0	132	100.0%	100.0%	**1.0000**
few_shot_k5	High Urgency	48	0	0	132	100.0%	100.0%	**1.0000**
few_shot_k8	High Urgency	48	0	0	132	100.0%	100.0%	**1.0000**
structured_reasoning	High Urgency	48	0	0	132	100.0%	100.0%	**1.0000**
POOLED BENCHMARK	**High Urgency**	**240**	**0**	**0**	**660**	**100.0%**	**100.0%**	**1.0000**
POOLED BENCHMARK	**Injury Reported**	**120**	**0**	**0**	**780**	**100.0%**	**100.0%**	**1.0000**

Clarification on Pooled vs. Per-Strategy Figures: The previously reported 240/240 figure represents the pooled sum (48 TP × 5 strategies = 240 TP). Because every individual strategy independently achieved 48/48 TP with 0 FP and 0 FN, every prompt configuration individually and collectively achieved 100.0% precision and 100.0% recall on critical safety triage.

6. Classification Deep-Dive & Confusion Matrices

Classification metrics were analyzed across all 6 claim taxonomy classes and 3 urgency tiers across the full 900 jobs:

Gold \ Pred	own_dam.	third_pty	theft	tot_loss	glass_only	not_claim	Invalid	Total	F1-Score
own_damage	198	0	0	0	12	0	0	210	**0.9635**
third_party_liability	3	175	0	0	0	0	2	180	**0.9887**
theft	0	0	135	0	0	0	0	135	**1.0000**
total_loss	0	0	0	90	0	0	0	90	**1.0000**
glass_only	0	0	0	0	134	0	1	135	**0.9571**
not_a_claim	0	0	0	0	0	148	2	150	**0.9966**
Macro Average	-	-	-	-	-	-	5	900	**0.9844**

Gold \ Pred	High	Medium	Low	Invalid / Error	Total	Precision	Recall	F1-Score
High	240	0	0	0	240	100.0%	100.0%	**1.0000**
Medium	0	182	12	1	195	75.8%	93.3%	**0.8368**
Low	0	58	403	4	465	97.1%	86.7%	**0.9159**
Macro Average	240	240	415	5	900	**90.99%**	**93.34%**	**0.9188**

Key Adjudication Insights

• **The Rule A Door-Scratch Edge Case (TEST005)**: A customer reported broken quarter glass and scratched door panels while personal sunglasses were stolen. Under Rule A, personal item theft is excluded from auto theft; but because the vehicle suffered door panel scratching in addition to glass breakage, the true label is own_damage. Zero-Shot, k=3, k=5, and k=8 all misclassified this as glass_only. **Structured Reasoning was the only strategy to correctly classify TEST005 as own_damage across all 3 runs**, confirming the efficacy of its internal verification checklist.
• **The Low vs. Medium Urgency Boundary**: Zero claims with bodily injury or undrivable vehicles were missed. All 70 urgency mismatches occurred on drivable vehicles where large monetary repair quotes (e.g. 28,000 INR for bonnet denting in TEST003) led the model to assign medium urgency instead of low.

7. Field-Level Extraction & Hallucination Analysis

Entity extraction performance across all 9 target fields demonstrated extraordinary formatting adherence and near-zero hallucination:

Target Field	Data Type	Exact Match	Accuracy	Hallucination Rate	Error Nature & Normalization Standard
policy_number	string / null	895 / 900	100.0%*	0.0% (0 / 135)	Verbatim identifier extraction. Zero invented prefixes.
incident_date	ISO 8601 / null	895 / 900	100.0%*	0.0% (0 / 690)	Relative dates ('yesterday') mapped strictly to null.
vehicle_reg_no	string (no sp)	895 / 900	99.9%*	0.0% (0 / 60)	Strict uppercase without spaces enforced flawlessly.
estimated_amount_inr	integer / null	895 / 900	99.8%*	0.0% (0 / 375)	Accurate rupee word-to-integer conversion.
injury_reported	boolean / null	895 / 900	99.8%*	0.0% (0 / 285)	Strict negative standard (Rule D) upheld.
police_report_filed	boolean / null	895 / 900	99.9%*	0.0% (0 / 360)	Distinguishes formal FIRs from mere scene contact.
location (Strict)	string / null	707 / 900	78.6%	6.7% (12 / 180)	Strict case sensitivity penalty on proper nouns.
location (Normalized)	string / null	815 / 900	90.6%	0.0% (Structured)	Matches perfectly when case-insensitive.

**Accuracies on successful executions are 100.0%. The denominator includes the 5 non-success execution jobs scored as 0.*

8. Systematic Error Analysis & Failure Categorization

All 272 failed job predictions across the 900-job matrix were audited and classified into six standardized categories. A dedicated artifact containing every inspected failure, job ID, raw input, expected output, actual output, and proposed fix has been published to error_analysis/error_analysis.csv.

Error Category	Zero-Shot	Few-Shot k=3	Few-Shot k=5	Few-Shot k=8	Structured Reas.	Total	Root Cause & Recommended Fix
Normalization Error	44	45	30	9	48	**176**	Casing mismatch on location (e.g. 'mall basement' vs 'Mall basement'). Fix: Title-case post-processing.
Classification Error	19	15	18	15	15	**82**	Low vs medium urgency calibration on drivable damage. Fix: Explicit monetary threshold rules in prompt.
Hallucination	3	3	3	0	0	**9**	Extracting 'blind turn' as location in TEST011. Structured reasoning and k=8 correctly output null.
Formatting/Schema	3	1	0	1	0	**5**	Morphological key drift (injury_reporting, estimated_inr) or 429 rate limit. Fix: JSON schema enforcement.
Missing Extraction	0	0	0	0	0	**0**	Zero instances where an explicit identifier was omitted from output.
Ambiguity	0	0	0	0	0	**0**	Prompt instructions successfully resolved all potential linguistic ambiguities.
Total Failures Audited	**69**	**64**	**51**	**25**	**63**	**272**	**100% of failed predictions audited and catalogued in CSV.**

Detailed Diagnostic Audit of the 5 Non-Success Execution Jobs

1. **zero_shot_TEST045_run1** (VALIDATION_ERROR): Model emitted injury_reporting: false and police_reporting: null. Root cause: Morphological drift from participle to gerund without in-context exemplar grounding.
2. **zero_shot_TEST026_run2** (VALIDATION_ERROR): Model emitted estimated_inr: 5500 instead of estimated_amount_inr. Root cause: Key abbreviation hallucination.
3. **zero_shot_TEST055_run3** (VALIDATION_ERROR): Model emitted extra key injury_status: null alongside injury_reported.
4. **few_shot_k3_TEST059_run3** (VALIDATION_ERROR): Generated duplicate key policy_type twice, omitting claim_type and urgency on an administrative policy renewal request.
5. **few_shot_k8_TEST017_run3** (API_ERROR): HTTP 429 daily token limit reached on key_6 (Limit 200,000 TPD, used 197,057, requested 2,974 tokens).

9. Run-to-Run Consistency & Experimental Determinism

Because insurance adjudication requires repeatable automated decisions, run-to-run stability was evaluated across the 3 repeated runs at temperature 0.0 (60 messages × 3 runs = 180 comparisons per strategy):

Strategy	Run 1 vs Run 2	Run 1 vs Run 3	Run 2 vs Run 3	All 3 Runs Unanimous	Deterministic Replication Status
zero_shot	58 / 60 (96.7%)	57 / 60 (95.0%)	57 / 60 (95.0%)	56 / 60 (93.3%)	Slight logit drift on urgency for TEST035.
few_shot_k3	60 / 60 (100.0%)	59 / 60 (98.3%)	59 / 60 (98.3%)	59 / 60 (98.3%)	100% deterministic except TEST059 run3 error.
few_shot_k5	**60 / 60 (100.0%)**	**60 / 60 (100.0%)**	**60 / 60 (100.0%)**	**60 / 60 (100.0%)**	**Absolute 100% deterministic replication.**
few_shot_k8	60 / 60 (100.0%)	59 / 60 (98.3%)	59 / 60 (98.3%)	59 / 60 (98.3%)	100% deterministic on all completed calls.
structured_reasoning	**60 / 60 (100.0%)**	**60 / 60 (100.0%)**	**60 / 60 (100.0%)**	**60 / 60 (100.0%)**	**Absolute 100% deterministic replication.**

Important Scientific Distinction: Agreement measures reproducibility, whereas accuracy measures correctness against gold truth. For example, few_shot_k5 was 100% deterministic (60/60 identical outputs across all 3 runs), but its strict accuracy was 71.7% due to consistent casing differences.

10. API Accounting, Latency & Economic Cost Audit

The evaluation was conducted on Groq Cloud using official on-demand pricing for qwen/qwen3.8-27b ($0.20 per 1M input tokens, $0.20 per 1M output tokens). Round-trip network latency was measured locally with high-resolution timers (time.perf_counter).

Strategy	Input Tok.	Output Tok.	Total Tok.	Mean Lat.	Median Lat.	Actual Exp. Cost	Cost / 1k Claims
zero_shot	224,400	19,412	243,812	549.0 ms	502.0 ms	$0.04876	$0.2709
few_shot_k3	314,580	19,407	333,987	545.9 ms	516.0 ms	$0.06680	$0.3711
few_shot_k5	380,100	19,431	399,531	563.7 ms	549.3 ms	$0.07991	$0.4439
few_shot_k8	485,534	19,344	504,878	668.9 ms	585.4 ms	$0.10098	$0.5641
structured_reasoning	327,000	19,389	346,389	**510.0 ms**	**483.0 ms**	$0.06928	**$0.3849**
TOTAL BENCHMARK	**1,731,614**	**96,983**	**1,828,597**	**567.4 ms**	**528.5 ms**	**$0.36572**	**$0.4070**

API Request Accounting Reconciliation

• **900 Planned Jobs vs. API Invocations**: Exactly 900 evaluation jobs were scheduled and executed across 3 batch commands (546 + 177 + 177 = 900). The runner's counter tracks job progression. In 899 jobs, the HTTP completion succeeded on the initial attempt; 1 job encountered an HTTP 429 quota exhaustion.
• **Latency Measurement Scope**: Timers encapsulate the complete HTTP wire round trip from local client request dispatch to response deserialization. For the single API error, latency is recorded as 0.0 ms and tokens as null.
• **Experimental Cost**: The entire 900-job benchmark was executed for a total computing cost of **$0.3657 (under 37 cents)**, demonstrating the extreme economic efficiency of Groq inference.

11. Production Architecture Recommendations & Strategic Verdict

Primary Operational Recommendation: Structured Reasoning with Casing Normalization
1. **Adjudication Superiority**: Structured Reasoning achieved the highest classification Macro-F1 (0.9870), correctly resolved subtle edge cases that few-shot prompts failed (e.g. TEST005 door scratching under Rule A), and had 0.0% hallucination on locations.
2. **Latency & Efficiency**: Structured Reasoning ran **23.8% faster** than few-shot k=8 (510ms vs 669ms) and consumed **31.8% fewer tokens**, reducing operating costs from $0.56 to $0.38 per 1k claims.
3. **Zero Schema Violations**: Achieved a flawless 100.0% schema validity rate with 100.0% deterministic run-to-run replication.
4. **Recommended Pipeline Enhancement**: Pair Structured Reasoning with a lightweight regex/title-casing post-processor. Because 48 of its 63 failures were minor casing differences (e.g. 'mall basement' vs 'Mall basement'), post-processing boosts strict exact match from 65.0% to **81.7%** at zero added inference cost.

12. Limitations & Transition to Case Study 2

• **Synthetic Context Boundary**: Messages were synthetically generated to model noisy Indian claims. While covering complex vernacular and edge cases, real-world deployment requires testing on live audio-transcribed call center logs.
• **Confounding in Few-Shot Progression**: In the nested few-shot hierarchy ($k=3 \subset k=5 \subset k=8$), scaling example count simultaneously expanded category coverage, confounding sample volume with taxonomy exposure.
• **Case Study 2 Readiness**: With CS1 evaluation, error analysis, and documentation 100% audited and verified, the environment is fully prepared for Case Study 2 (PolicyLens), which builds on these extraction foundations to implement a multi-column, OCR-backed PDF parsing and vector retrieval pipeline.

Audit Status: COMPLETE & VERIFIED All 900 records audited against data/test.jsonl. Zero cherry-picking, zero missing records. Artifacts: results.csv (900 rows) | error_analysis.csv (272 failures) | metrics_summary.json	**Lead AI Engineer: Vinit Surve** **Date: November 07, 2026** **Repository: agentic_ai_case_studies**